

# The Psychology of Data Visualization

Fall 2026 ■ MWF 3:35-4:25p

2026-07-16

Spring 2025 ■ TTh 9:05-10:25 AM



Figure 1: ChatGPT/Dall-E-3 model response to ‘an image of “a picture paints a thousand words” ’ on 2024-09-19

## Themes

If a picture’s worth a thousand words (Wikipedia contributors 2024a), what exactly does it say? This course will focus on the psychology of data visualization—how to read, critique, and generate meaningful figures that inform but don’t mislead. We’ll take inspiration from recognized classic figures and unpack what makes them exemplary. We’ll critique figures that

deliberately or inadvertently mislead or confuse. We'll learn what 150 years of vision science, the foundation of experimental psychology, has to say about data visualization. And we'll learn how to build our own reproducible figures using Python, R, and JavaScript. No prior programming experience is required.

### **Instructor**

Rick O. Gilmore, Ph.D.  
Professor of Psychology  
rog1 AT-SIGN psu PERIOD edu

### **Office hours**

Monday & Friday, 12:00p - 1:00p in 114 Moore or by appointment: <https://doodle.com/mm/rickgilmore/book-a-time>

Lab web site: <https://gilmore-lab.github.io>

### **Teaching Assistant**

Kalvin Garrah  
Graduate Student in Social Psychology  
kmg7035 AT-SIGN psu PERIOD edu

### **Office hours**

Tuesday, 12:00p - 2:00p, in 473 Moore or by appointment (email).

### **Meeting time & location**

Monday, Wednesday, & Friday, 3:35 pm - 4:25 pm  
216 [Osmond Lab](#)

## Canvas site

Most of the course content will be found on this site.

However, we will use Canvas to track attendance, submit and grade assignments, and provide ongoing summaries about grade point totals. The Canvas site may be found here:

<https://psu.instructure.com/courses/2484925>

## Course structure

This is a discussion-focused course. On most days we will discuss readings assigned prior to class. On many days, we will work together or individually on the assigned [exercises](#), the final project, or another assignment. Please remember to bring your personal laptop computer. If you do not have a personal laptop, contact Dr. Gilmore so that we can make arrangements for you to participate fully in the course activities.

## Week 01

*Surveying the landscape*

### Mon, Aug 24

*Course introduction*

- *Watch:*
  - PBSoffbook (2013)
- [Slides](#)

### Wed, Aug 26

*The semiotics of data visualization*

- *Read:*
  - Cairo (2013), Chapter 1-2, [PDF on Canvas](#)
- *Assigned:* [Exercise 01](#) assigned
- [Slides](#)
- *Homework:*
  - Install R and [RStudio Desktop](#)

- \* *Tutorial*
- (Optional) Install [Quarto](#)

## Fri, Aug 28

*Work Session: Introducing R & RStudio*

- *Complete:* R and RStudio Desktop installation and setup
- *Read:* [Get Started](#) from the RStudio User Guide.
- *Skim:*
  - <https://quarto.org/docs/get-started/hello/jupyter.html><sup>1</sup>
  - <https://education.rstudio.com/learn/beginner/>
- [Slides](#)

## Week 02

*Who visualizes data and why*

## Mon, Aug 31

*Visualization in government & business*

- [Slides](#)

## Wed, Sep 02

*Visualization in art, sports, and journalism*

- *Read*
  - Gelman and Unwin (2013) (recommended), follow citation link to online version or [PDF on Canvas](#).
- *Assigned:* [Exercise 02](#)
- [Slides](#)
- [Results](#) from [Exercise 01](#)

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<sup>1</sup>We won't use Python in this course, but the basic ideas apply to Python and R.

## Fri, Sep 04

*Work Session: Exercise 02 and Quarto*

- 📅 Due [Exercise 01](#) | 📁 [Canvas dropbox](#)
- [Slides](#)

## Week 03

*Understanding figures*

## Mon, Sep 07

**NO CLASS: Labor Day**

## Wed, Sep 09

*Making (sense of) data*

- *Read:*
  - Stevens (1946) | [PDF on Canvas](#)
  - Wikipedia contributors (2024b) (Optional)
  - Tourangeau et al. (2012) | [PDF on Canvas](#). (Optional)
- [Slides](#)

## Fri, Sep 11

*Work Session: Making data with R*

- [Slides](#)

## Week 04

*The anatomy of figures*

## Mon, Sep 14

### *Figure types*

- *Explore* (pick at least two):
  - Rebecca (n.d.)
  - Hammond (2024)
  - Discovery (2024)
  - Atlassian (n.d.)
- *Watch*:
  - Channel (2024)
- Assigned: [Exercise-03](#)
- [Slides](#)

## Wed, Sep 16

### *More figure types and figure components*

- Read: [Tutorial: Making data](#)
- [Slides](#)

## Fri, Sep 18

### *Work Session: Intro to ggplot*

- [Slides](#)
- [ggplot2 tutorial](#)
- Continue work on [Exercise 02](#) and [Exercise 03](#).

## Week 05

### *Perceptual foundations*

## Mon, Sep 21

NO CLASS

## Wed, Sep 23

*From stimulus to sensation*

- *Read:*
  - Cairo (2013), Chapter 5, [PDF on Canvas](#)
- [Slides](#)

## Fri, Sep 25

*Work Session: More ggplot*

- 📅 Due [Exercise 02](#) | 📁 [Canvas dropbox](#) |
- [Tutorial](#)

## Week 06

*Visual perception and cognition*

## Mon, Sep 28

*From sensation to perception*

- *Read:*
  - Cairo (2013), Chapter 6, [PDF on Canvas](#)
  - Few (2004), Chapter 5, [PDF on Canvas](#)
- *Optional:*
  - Cleveland and McGill (1984)
- [Slides](#)

## Wed, Sep 30

*From cognition to understanding*

- *Read:*
  - Franconeri et al. (2021), pp. 109-122, [PDF on Canvas](#)
  - Cairo (2013), Chapter 7, [PDF on Canvas](#)
- Assigned: [Final Project](#) proposal
- [Slides](#)

## Fri, Oct 02

*Work Session: Visualizing sports data*

- 📅 Due [Exercise-03](#) | 📁 *Canvas dropbox* ](<https://psu.instructure.com/courses/2484925/assignments/18448437>) |
- [Tutorial](#)
- [Slides](#)

## Week 07

### Mon, Oct 05

*Exploring data*

- *Read:*
  - Tukey (1977), Chapter 1, [PDF on Canvas](#)
- [Slides](#)

### Wed, Oct 07

*Communicating uncertainty and risk*

- *Read:*
  - Franconeri et al. (2021), pp. 139-150, [PDF on Canvas](#)
  - Wilke (2019), Ch. 16, [PDF on Canvas](#)
- [Slides](#)

## Fri, Oct 09

*Work Session: More sports data viz*

- 📅 Due [Exercise-04](#) | 📁 [Canvas dropbox](#) |
- [Slides](#)

## Week 08

### Mon, Oct 12

*Critiquing figures*

- \*Read\*:
  - Tufte (2001), Chapter 1, [PDF on Canvas](#)
  - Tufte (2001), Chapter 2, [PDF on Canvas](#)
- [Slides](#)

### Wed, Oct 14

*Storytelling with data*

- *Read*:
  - Knafllic (2015), pp. ix-33, [PDF on Canvas](#)
  - Knafllic (2015), pp. 33-69, [PDF on Canvas](#)
- *Skim*: Woodside et al. (2008), [PDF on Canvas](#)
- Assigned: [Exercise-05](#)
- [Slides](#)

## Fri, Oct 16

*Work Session: TidyTuesday or Exercise 05*

- 📅 Due [Final Project](#) proposal | 📁 [Canvas dropbox](#)
- [Slides](#)

## Week 09

*Visualizing spatial data*

## Mon, Oct 19

### *Map-making*

- Read:
  - Wilke (2019), Chapter 15, [PDF on Canvas](#)
- [Slides](#)

## Wed, Oct 21

- [Slides](#)

## Fri, Oct 23

### *Work Session: TidyTuesday II*

- [Slides](#)

## Week 10

### *Turning the table(s)*

## Mon, Oct 26

### *Tables in Base R*

- 📅 Due [Exercise-05](#)
- [Slides](#)

## Wed, Oct 28

### *gt and knitr::kable*

- [Slides](#)

**Fri, Oct 30**

*Work Session: Tables*

- [Slides](#)

**Week 11**

*Data viz in policy & politics*

**Mon, Nov 02**

- [Slides](#)

**Wed, Nov 04**

- [Slides](#)

**Fri, Nov 06**

**Week 12**

**Mon, Nov 09**

**Wed, Nov 11**

**Fri, Nov 13**

*Work Session: More TidyTuesday*

- [Slides](#)

**Week 13**

*Data viz and quantitative inference*

## **Mon, Nov 16**

*Synthesize before you analyze*

- [Slides](#)
- 📅 Due [Exercise-06](#)

## **Wed, Nov 18**

*Sample size, effect size, statistical power*

- [Slides](#)

## **Fri, Nov 20**

*Work Session: Final projects*

## **Thanksgiving Break**

**NO CLASS**

## **Week 14**

## **Mon, Nov 30**

*Making data dashboards and apps*

- Explore:
  - Kohler (2024)
- [Slides](#)

## **Wed, Dec 02**

*Work Session: Final projects*

## **Fri, Dec 04**

*Work Session: Final projects*

## Week 15

Mon, Dec 07

*Final Project Presentations*

Wed, Dec 09

*Final Project Presentations*

Fri, Dec 11

*Final Project Presentations*

## Finals Week

Wed, Dec 16

- 📅 Due [Final project](#) write-up | 📁 [Canvas dropbox](#)

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## Elements

| Component     | Description                                                                                                                                          | Points |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Attendance    | You will receive 1 point for each class you attend up to a maximum of 40.                                                                            | 40     |
| Exercises     | There will be six (6) exercises that we will work on. Each exercise is worth 10 points. You are required to submit four (4) toward your final grade. | 40     |
| Final project | You will complete a <a href="#">final project</a> , either on your own, or with a small group of 3 or fewer. Your final project is worth 40 points.  | 40     |
|               | <b>TOTAL POINTS POSSIBLE</b>                                                                                                                         | 120    |
| Extra Credit  | If you submit more than four (4) exercise write-ups, you may earn up to 10 extra credit points for these exercises.                                  |        |

## Grading Scheme

| Percent | Points  | Grade |
|---------|---------|-------|
| 94+     | 113+    | A     |
| 90-93   | 108-112 | A-    |
| 87-89   | 104-107 | B+    |
| 84-86   | 100-103 | B     |
| 80-83   | 96-100  | B-    |
| 77-79   | 92-95   | C+    |
| 70-76   | 84-91   | C     |
| 60-69   | 72-83   | D     |
| <59     | <=71    | F     |

This page summarizes some of the key deadlines in the course.

| Date                       | What's due/happening                      |
|----------------------------|-------------------------------------------|
| <a href="#">2026-09-04</a> | <a href="#">Exercise 01</a>               |
| <a href="#">2026-09-25</a> | <a href="#">Exercise 02</a>               |
| <a href="#">2026-10-02</a> | <a href="#">Exercise 03</a>               |
| <a href="#">2026-10-09</a> | <a href="#">Exercise 04</a>               |
| <a href="#">2026-10-16</a> | <a href="#">Final project proposal</a>    |
| <a href="#">2025-10-26</a> | <a href="#">Exercise 05</a>               |
| <a href="#">2026-11-16</a> | <a href="#">Exercise 06</a>               |
| <a href="#">2026-12-16</a> | <a href="#">Final project</a> writeup due |

Figure [Figure 2](#) below shows one way to visualize these deadlines. We'll examine it and attempt to improve it at some point in the class.

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

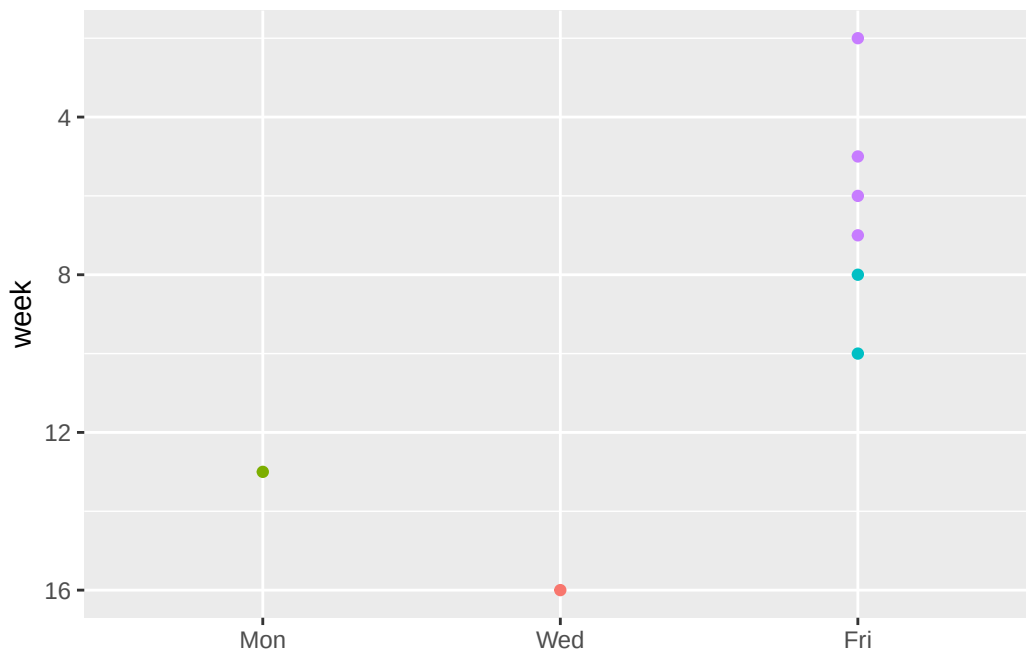


Figure 2: Visualization of assignment deadlines in PSYCH 490.007 Fall 2026 by week of the semester, day of the week, and month.

## Academic Integrity

Students with questions about academic integrity should visit <http://www.la.psu.edu/current-students/undergraduate-students/education/academic-integrity>.

Penn State defines academic integrity as the pursuit of scholarly activity in an open, honest and responsible manner. All students should act with personal integrity, respect others dignity, rights and property, and help create and maintain an environment in which all can succeed through the fruits of their efforts (Faculty Senate Policy 49-20). Sanctions for academic misconduct can include a grade of F for the course as well as other penalties.

Unless you are told otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and you may not assist other students with papers, quizzes, exams, or other assessments. If I allow you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source.

When you complete assignments, remember the **ABCs** to avoid plagiarism: **A**lways place copied information within quotation marks, include information about the quoted or paraphrased source in a **B**ibliography, and **C**ite the source in the body (in the text) of your paper immediately

after the quoted or paraphrased information. When in doubt, cite in the text and include the source in a bibliography.

**Students with questions about academic integrity should ask me or the TA before submitting work.**

Students facing allegations of academic misconduct may not drop/withdraw from the affected course unless they are cleared of wrongdoing (see G-9: Academic Integrity). Attempted drops will be prevented or reversed, and students will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Dean's List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

### **Use of Artificial Intelligence (AI)**

You may use AI tools to assist with your work, but you must properly cite any sources used. This includes AI-generated content, such as text, images, data, or code.

Always ensure that your work reflects your own understanding and that you are not relying solely on AI tools to complete assignments.

If you do make use of AI, please cite the AI tool used and provide a brief description of how it was used in your work. Please also save a transcript of your interaction with the AI tool and submit that as an appendix to your assignment.

See <https://integrity.psu.edu/ai> for more details, and when in doubt, please ask the instructor or TA.

### **Absences or late assignments**

#### **Absence from class**

Your absence from class may be excused under unusual circumstances such as (a) an interview for graduate school or a job, (b) illness, (c) religious observance, (d) the death of a family member, or (e) any other event recognized by the university as a valid excuse for absence from class.

If you must miss class, you must contact the instructor and the TA **in advance**.

Up to three (3) excused absences will be permitted.

### **Late exercises**

Exercises submitted after the published deadlines will not be eligible for full credit unless the instructor has given specific permission.

### **Late final projects**

Final projects submitted after the published deadline will not be eligible for full credit unless the instructor has given specific permission.

### **Accommodation for persons with disabilities**

Penn State welcomes students with disabilities into the University's educational programs. Please refer to the information provided by Student Disability Resources (SDR) at <http://equity.psu.edu/student-disability-resources/> for information about the procedures required to obtain reasonable accommodations in this course. Students should discuss SDR-approved accommodations with their instructor as early in the semester as possible, even if they have taken another course with the instructor. Please note: students are not required to provide their instructor with information about the nature of their condition.

Penn State students are also welcome to contact other units for assistance with personal concerns that interfere with academic progress, including: Counseling and Psychological Services (CAPS; <http://studentaffairs.psu.edu/counseling/>), the Office of Student Affairs (<http://studentaffairs.psu.edu/>), Career Services (<http://studentaffairs.psu.edu/career/>), the Center for Women Students (<http://studentaffairs.psu.edu/womenscenter/>), the LGBTQA Student Resource Center (<http://studentaffairs.psu.edu/lgbtqa/>), the Office of Sexual Misconduct Prevention and Response (<http://titleix.psu.edu/>), Penn State Educational Equity (<http://equity.psu.edu/>), the Multicultural Resource Center (<http://equity.psu.edu/mrc/>), and University Health Services (<http://studentaffairs.psu.edu/health/>).

### **Nondiscrimination Statement**

The Pennsylvania State University is committed to equal access to programs, facilities, admission and employment for all persons. It is the policy of the University to maintain an environment free of harassment and free of discrimination against any person because of age, race, color, ancestry, national origin, religion, creed, service in the uniformed services (as defined in state and federal law), veteran status, sex, sexual orientation, marital or family status, pregnancy, pregnancy-related conditions, physical or mental disability, gender, perceived gender, gender identity, genetic information or political ideas.

Discriminatory conduct and harassment, as well as sexual misconduct and relationship violence, violates the dignity of individuals, impedes the realization of the University's educational mission, and will not be tolerated.

Direct all inquiries regarding the nondiscrimination policy to:

Suzanne C. Adair Associate Vice President for Equal Opportunity and Access Office of Equal Opportunity and Access The Pennsylvania State University 328 Boucke Building University Park, PA 16802-5901 Email: sca917@psu.edu Tel (814) 863-0471

### **Diversity Statement**

This classroom is a place where you will be treated with respect. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class.

Penn State is committed to creating an educational environment which is free from intolerance directed toward individuals or groups and strives to create and maintain an environment that fosters respect for others as stated in Policy AD29 Statement on Intolerance.

### **Mandated Reporting Statement**

Penn State's policies require me, as a faculty member, to share information about incidents of sex-based discrimination and harassment (discrimination, harassment, sexual harassment, sexual misconduct, dating violence, domestic violence, stalking, and retaliation) with Penn State's Title IX coordinator or deputy coordinators, regardless of whether the incidents are stated to me in person or shared by students as part of their coursework. For more information regarding the University's policies and procedures for responding to reports of sexual or gender-based harassment or misconduct, please visit <http://titleix.psu.edu>.

Additionally, I am required to make a report on any reasonable suspicion of child abuse in accordance with the Pennsylvania Child Protective Services Law.

### **Zoom**

At some point in the semester, I may decide to use Zoom to allow students who are unable to attend class in person to participate.

While you are on Zoom, keep in mind that this is a classroom environment and others should be treated with respect. Please keep your microphone muted unless you want to ask a question or interact with someone. If your microphone is not muted, the entire class will be able to hear what is going on in your environment. As an instructor, I personally like to see people's faces. As a participant, I am more involved when I have my camera on. I realize, however, that there are many reasons why you might not want to turn on your camera such as poor internet connection, joining via phone, or other privacy concerns. It is your choice as to whether you would like to have the camera on or not.

## Values

### Penn State Principles

The Pennsylvania State University is a community dedicated to personal and academic excellence. The Penn State Principles were developed to embody the values that we hope our students, faculty, staff, administration, and alumni possess. At the same time, the University is strongly committed to freedom of expression. Consequently, these Principles do not constitute University policy and are not intended to interfere in any way with an individual's academic or personal freedoms. We hope, however, that individuals will voluntarily endorse these common principles, thereby contributing to the traditions and scholarly heritage left by those who preceded them, and will thus leave Penn State a better place for those who follow.

**I will respect the dignity of all individuals within the Penn State community.**

The University is committed to creating and maintaining an educational environment that respects the right of all individuals to participate fully in the community. Actions motivated by hate, prejudice, or intolerance violate this principle. I will not engage in any behaviors that compromise or demean the dignity of individuals or groups, including intimidation, stalking, harassment, discrimination, taunting, ridiculing, insulting, or acts of violence. I will demonstrate respect for others by striving to learn from differences between people, ideas, and opinions and by avoiding behaviors that inhibit the ability of other community members to feel safe or welcome as they pursue their academic goals.

**I will practice academic integrity.** Academic integrity is a basic guiding principle for all academic activity at Penn State University, allowing the pursuit of scholarly activity in an open, honest, and responsible manner. In accordance with the University Code of Conduct, I will practice integrity in regard to all academic assignments. I will not engage in or tolerate acts of falsification, misrepresentation or deception because such acts of dishonesty violate the fundamental ethical principles of the University community and compromise the worth of work completed by others.

**I will demonstrate social and personal responsibility.** The University is a community that promotes learning; any behaviors that are inconsistent with that goal are unacceptable. Irresponsible behaviors, including alcohol or drug abuse and the use of violence against people or property, undermine the educational climate by threatening the physical and mental health of members of the community. I will exercise personal responsibility for my actions and I will make sure that my actions do not interfere with the academic and social environment of the University. I will maintain a high standard of behavior by adhering to the Code of Conduct and respecting the rights of others.

**I will be responsible for my own academic progress and agree to comply with all University policies.** The University allows students to identify and achieve their academic goals by providing the information needed to plan the chosen program of study and the necessary educational opportunities, but students assume final responsibility for course

scheduling, program planning, and the successful completion of graduation requirements. I will be responsible for seeking the academic and career information needed to meet my educational goals by becoming knowledgeable about the relevant policies, procedures, and rules of the University and academic program, by consulting and meeting with my adviser, and by successfully completing all of the requirements for graduation.

## **Penn State Values**

**Integrity:** We act with integrity and honesty in accordance with the highest academic, professional, and ethical standards.

**Respect:** We respect and honor the dignity of each person, embrace civil discourse, and foster a diverse and inclusive community.

**Responsibility:** We act responsibly, and we are accountable for our decisions, actions, and their consequences.

**Discovery:** We seek and create new knowledge and understanding, and foster creativity and innovation, for the benefit of our communities, society, and the environment.

**Excellence:** We strive for excellence in all our endeavors as individuals, an institution, and a leader in higher education.

**Community:** We work together for the betterment of our University, the communities we serve, and the world.

## **References**

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